

19. Study the sequence of bases in a DNA molecule that is shown below and answer the questions.

5' – GAATTC 3'

3' – CTTAAG – 5'

- (a) What are such sequences called ? Name the enzyme used that recognises such nucleotide sequences. 1
- (b) How do these enzymes function ? 1
20. (a) Analyse the following two cases of sex determination in different organisms :
- Case I : Males have (XO) sex chromosomes and females have two copies of same sex chromosome (XX).
- Case II : Females have two different sex chromosomes (ZW) and males have two copies of same sex chromosome (ZZ).
- Identify the type of heterogamety in each case, giving one example of each. 1 + 1

OR

- (b) (i) Differentiate between a DNA and RNA nucleotide. 1
- (ii) Name the two types of bonds present within a single nucleotide. 1
21. (a) Given below is a pyramid found in an ecosystem, where each bar represents the standing crop available in the trophic level.



- (i) Identify the kind of pyramid and with the help of an example explain the conditions where this kind of pyramid is possible in nature. 1
- (ii) Write any two limitations of ecological pyramids. 1

OR

- (b) In an ecosystem there was loss of biodiversity due to some project in that area.
- (i) How will biodiversity be affected ? (2 points) 1
- (ii) List two major causes of loss of biodiversity. 1